

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

The UDP protocol reveals that the DNS requests sent to the DNS server are not being completed. This is based on the results of the network analysis, which show that the ICMP echo reply returned the error message "UDP Port unreachable". The port noted in the error message is used for DNS traffic, which translates domain names into IP addresses. The most likely issue is the UDP message request did not go through to the DNS server because no service was listening on the receiving DNS port.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

This incident occurred at 1:24. Our IT team became aware of the incident due to customers reporting they were unable to reach the client company website. The IT team has taken investigative measures attempting to access the website using tcpdump to analyze the DNS and ICMP network traffic. DNS requests were sent to the DNS server using UDP port 53, but the server returned ICMP messages stating the port was unreachable. This prevented the domain name from being translated into an IP address. The cause of this event is more than likely because the port is blocked.